Assignment No-5

Q.1 Count digit:
n = 54321. count = 0

n % 10 = 1
Count ++ = 1
n/10 = 5432.

n%10 = 2 n = 5432.
Count++ = 2
n/10 = 543

n%10 = 3 n = 543
Count ++ = 3
n/10 = 54

n%10 = 4 n = 54
Count++ = 4
n/10 = 5

n%10 = 5 n = 5
Count++ = 5
n/10 = 0

$\boxed{\text{Count} = 5}$

```
int n = 54321;
int count = 0;
while (n != 0) {
    int rem = n % 10;
    Count++;
    n = n/10;
}

Sop (count);
```

2) extract digits.
n = 5328;

n = 5328 n%10 = 8 = rem.
print (rem) = 8
n/10 = 532.

n = 532 n%10 = rem = 2
print = 2
n/10 = 53

n = 53 n%10 = 3 = rem
print = 3
n/10 = 5

n = 5 ~~print~~ n%10 = 5 = rem
print (5)
n/10 = 0

o/p = 8 2 3 5

```
int n = 5328
while (n != 0) {
    int rem = n%10;
    System.out.println (rem);
    n = n/10;
}
```

o/p : 8
2
3
5

TC = (log n) = n/10 - divide.
SC ~~(~~ O(1)

③ Reverse ano:

$n = 12345$, $rev = 0$

$n = 12345$ $\quad n \% 10 = rem = 5$

$rev = rev * 10 + rem$

$= 0 * 10 + 5 = 5$

$n / 10 = 1234$

$n = 1234$ $\quad rev = 5$

$n \% 10 = 4$

$rev = 5 * 10 + 4 = 54$

$n / 10 = 123$

$n = 123$ $\quad rev = 54$

$n \% 10 = 3$

$rev = 54 * 10 + 3 = 543$

$n / 10 = 12$

$n = 12$ $\quad rev = 543$

$n \% 10 = 2$

$rev = 543 * 10 + 2 = 5432$

$n / 10 = 1$

$n = 1$ $\quad rev = 5432$

$n \% 10 = 1$

$rev = 5432 * 10 + 1 = 54321$

$n / 10 = 0$

$TC = O(\log n) = n / 10 =$ divide.

$SC = O(1)$

```
int n = 12345;
int rev = 0;
while (n != 0){
        int rem = n%10;
        rev = rev * 10 + rem;
        n = n / 10;
}
Sop (rev);                    o/p = 54321
```

4] <u>palindrome no</u>: if reverse & orignal no is same then it is palindrome or not palindrome.

n = 121

n = 121 rev = 0
n%10 = 1
rev = rev * 10 + rem = 0 * 10 + 1 = 1
n/10 = 12

n = 12 rev = 1
n%10 = 2
rev = 2 * 10 + 1 = 21
n/10 = 1

n = 1 rev = 21
n%10 = 1
rev = 21 * 10 + 1 = 121
n/10 = 0.

orignal = 121 rev = 121
if (orignal == rev) = palindrome.

```
int n = 121;
int orignal = n;
int rev = 0;
while (n != 0){
    int rem = n % 10;
    rev = rev * 10 + rem;
    n = n / 10;
}
  if (orignal == rev){
        Sop("palindrome");
  }else{
        Sop("not palindrome");
}
```

o/p = palindrome.

O(log n) = TC ~~10/10~~ n/10 - divide.
O(1) = SC

Sum of digit = exp: 543 = 12. = 5+4+3.

n = 543 Sum = 0.
n%10 = 3
Sum = Sum + rem = 0+3 = 3
n/10 = 54

n = 54 Sum = 3
n%10 = 4
Sum = 3+4 = 7
n/10 = 5

n = 5 Sum = 7
n%10 = 5
Sum = 7+5 = 12
n/10 = 0

```
int n = 543;
int sum = 0;
While (n>0){
    int rem = n%10;
    Sum = Sum + rem;
    n=n/10;
}
    Sop (sum);
```

o/p = 12.

TC = O(log n) = n/10 divide
SC = O(1)

6) Armstrong no :

$n = 153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

int temp = n ;

153 != 0

└ Count ++ = 1

temp / 10 = 15

15 != 0

└ Count ++ = 2

temp / 10 = 1

1 != 0

└ Count ++ = 3

temp / 10 = 0

temp = n = temp = 153.

153 != 0 power = 1

└ 153 % 10 = 3

Sum = Sum + rem

0 = 0 + 27 = 27

i = 1 <= 3 — 1 = 1 * 3 = 3

i = 2 <= 3 — 3 = 3 * 3 = 9 = 27

i = 3 <= 3 — 9 = 9 * 3 = 27

153 / 10 = 15

15 != 0 power = 1

└ 15 % 10 = 5

27 + 125 = 152

i = 1 <= 3 — 1 = 1 * 5 = 5

i = 2 <= 3 — 5 = 5 * 5 = 25 = 125

i = 3 <= 3 — 25 = 25 * 5 = 125

15 / 10 = 1

11 = 0 power = 1

1 % 10 = 1

152 + 1 = 153.

i = 1 <= 3 — 1 = 1 * 1 = 1

i = 2 <= 3 — 1 = 1 * 1 = 1 = 1

i = 3 <= 3 — 1 = 1 * 1 = 1

if (27 + 125 + 1 == n) ÷ armstrong no.

```
int n = 153;
int temp = n;
int sum = 0;
int count = 0;
while (temp != 0){
        count ++;
        temp = temp / 10;
  }
  temp = n;
while (temp != 0){
        int rem = temp % 10;
          int power = 1;
          for (int i = 1; i <= count; i++){
                power = power * rem;
          }
          Sum = sum + power;
          temp = temp / 10;
  }
    if (sum == n){
                Sop ("armstrong");
    }else {
                Sop ("not armstrong");
```

7) print all divisors :

$n = 12$; $\sqrt{12} = 3.--$

$i = 1$ $12 \% 1 == 0$ $n/i = 12/1 = 12$

1

$i = 2$ $12 \% 2 == 0$ $n/i = 12/2 = 6$

2

$i = 3$ $12 \% 3 == 0$ $12/3 = 4$

3

So factors are : 1, 2, 3, 4, 6, 12.

```
int n = 12;
for (int i = 1; i <= Math.sqrt(n); i++){
        if (n % i == 0) {
                Sop (i);
                Sop (n/i);
        }
}
```

o/p = 1, 2, 3, 4, 6, 12.

$TC = O(\sqrt{n})$

$SC = O(1)$

8) Count no divisors.

$n = 12$; Count = 0 $\sqrt{12} = 3$

$i = 1$ $12 \% 1 == 0$ = Count = 1
$12 / 1 != 1$ = Count = 2

$i = 2$ $12 \% 2 == 0$ = Count = 3
$12 / 2 != 2$ = Count = 4

$i = 3$ $12 \% 3 == 0$ = Count = 5
$12 / 3 != 3$ = Count = 6.

So count = 6.

```
int n = 12;
int count = 0;
for (int i = 1; i <= Math.sqrt(n); i++){
        if (n % i == 0) {
                Count ++;
                if (i != n / i){
                        Count ++;
                }
        }
Sop (count) = 6.
```

o/p = 6.

TC = $O(\sqrt{n})$
SC = $O(1)$

9) prime no = 7 - prime 9 - not prime.

n = 7 = i <= n Count = 0.

i = 1 7 % 1 == 0 ✓ Count = 1

i = 2 7 % 2 == 0 ✗

i = 3 7 % 3 == 0 ✗

i = 4 7 % 4 == 0 ✗

i = 5 7 % 5 == 0 ✗

i = 6 7 % 6 == 0 ✗

i = 7 7 % 7 == 0 ✓ Count = 2

if (count == 2) { - prime no.

n = 12

i = 1 12 % 1 == 0 Count = 1

i = 2 12 % 2 == 0 Count = 2

i = 3 12 % 3 == 0 Count = 3

i = 4 12 % 4 == 0 Count = 4

i = 5 12 % 5 == 0 ✗

i = 6 12 % 6 == 0 Count = 5

i = 7 12 % 7 == 0 ✗

i = 8 12 % 8 == 0 ✗

i = 9 12 % 9 == 0 ✗

i = 10 12 % 10 == 0 ✗

i = 11 12 % 11 == 0 ✗

i = 12 12 % 12 == 0 Count = 6

Count == 6 = not prime.

```
int n=Y;
int count=0
for (int i=1; i<=n; i++){
        if (n%i==0) {
                count ++;
        }
}
  if (count==2){
          Sop ("prime no");
  } else{
          Sop ("not prime");
}
```

o/p = prime no.

TC = O(n) = i=1 to n.
SC = O(1)

10] Print all prime no : n = 20 if count == 2 - prime.
2, 3, 5, 7, 11, 13, 17, 19.

i = 1 1<=20
└ j=1<=1 - 1%1 == 0 Count = 1 - not pr

i=2 2<=20
└ j=1<=2 - 2%1 == 0 Count = 1
j=2<=2 - 2%2 == 0 Count = 2 - prime.

i=3 3<=20
└ j=1<=3 - 3%1 == 0 Count = 1
j=2<=3 - 3%2 == 0 X = prime
j=3<=3 = 3%3 == 0 Count = 2

i=4 4<=20
└ j=1<=4 - 4%1 == 0 Count = 1
j=2<=4 - 4%2 == 0 2 not prime.
3<=4 - 4%3 == 0 X
4<=4 - 4%4 == 0 3

i=5 5<=20
└ j=1<=5 - 5%1 == 0 ~~Count~~ Count = 1
2<=5 - 5%2 == 0 X
3<=5 - 5%3 == 0 X prime no.
4<=5 - 5%4 == 0 X
5<=5 - 5%5 == 0 = 2

i=6 6<=20
└ j=1<=6 - 6%1 == 0 Count = 1
2<=6 = 6%2 == 0 Count = 2

3<=6 - 6%3 == 0 - Count = 3
4<=6 - 6%4 == 0 ✗
5<=6 - 6%5 == 0 ✗ not prime.
6<=6 - 6%6 == 0 Count = 4

i=7 <=20
└ j=1 <=7 = 7%1 == 0 Count = 1
2<=7 = 7%2 == 0 ✗
3<=7 - 7%3 == 0 ✗
4<=7 - 7%4 == 0 ✗ prime no.
5<=7 - 7%5 == 0 ✗
6<=7 - 7%6 == 0 ✗
7<=7 - 7%7 == 0 Count = 2

```
int n=20;
for (int i=1; i<=n; i++){
    int count=0;
    for (int j=1; j<=i; j++){
        if (i%j==0) {
            count++;
        }
    }
    if (count==2){
        Sop (i);
    }
}
```

O/p : 2, 3, 5, 6, 7, 11, 13, 17, 19.

ii) find GCD - brute force.

a = 12;
b = 18;

min value

i = 1 <= 12
└ 12 % 1 == 0 && ~~factor = 1~~ ✓ gcd = 1
18 % 1 == 0

i = 2 <= 12
└ 12 % 2 == 0 && ✓ gcd = 2
18 % 2 == 0

i = 3 <= 12
└ 12 % 3 == 0 && ✗
18 % 3 == 0

i = 4 <= 12
└ 12 % 4 == 0 && ✗
18 % 4 == 0

i = 5 <= 12
└ 12 % 5 == 0 ✗

i = 6 <= 12
└ 12 % 6 == 0 ✓ gcd = 6
18 % 6 == 0

i = 7 <= 12
└ 12 % 7 == 0 ✗

i = 8 <= 12
└ 12 % 8 == 0 ✗

i=9 <=12
└ 12%9==0 ✗

i=10<=12
└ 12%10==0 ✗

i=11<=12
└ 12%11==0 ✗

i=12<=12
└ 12%12==0 ✗
18%12==0

gcd = 6

```
int a = 12;
int b = 18;
int gcd = 1;
for (int i=1; i<=Math.min(a,b); i++){
        if (a%i==0 && b%i==0){
                gcd=i;
        }
    }
    Sop (gcd);
}
```

O/p = 6

TC = O(min(a,b))
SC = O(1)

n) GCD using euclidean algorithm.

Ans: $a = 48$

$b = 18$

if $(a > b) = gcd(a, b) = gcd(a \% b, b)$

$gcd(48, 18) = gcd(48 \% 18, 18) = 12$

$gcd(12, 18) = gcd(18 \% 12, 18) = 6$

$gcd(12, 6) = gcd(12 \% 6, 6) = 0$

$gcd(6, 0)$

$o/p = 6$

```
int a = 48 ;
int b = 18 ;
int gcd = 1 ;
while (a != 0 && b != 0) {
        if (a > b) {
                a = a % b;
        }else{
                b = b % a;
        }
}
    if (a == 0) {
            gcd = b;
    }else{
            gcd = a;
}
```

$TC = O(\log n)$

$o/p = 6$

13) check coprime no :

a = 8 = min

b = 15

i = 1 <= 8

└ 8 % 1 == 0

15 % 1 == 0 gcd = 1

i = 2 <= 8

└ 8 % 2 == 0

15 % 2 == 0 ✗

i = 3 < 8

└ 8 % 3 == 0 ✗

i = 4 <= 8

└ 8 % 4 = 0 ✓

15 % 4 = 0 ✗

i = 5 <= 8

└ 8 % 5 == 0 ✗

i = 6 <= 8

└ 8 % 6 == 0 ✗

i = 7 <= 8

└ 8 % 7 == 0 ✗

i = 8 <= 8

└ 8 % 8 == 0 ✓

8 % 15 == 0 ✗

gcd = 1
if gcd = 1 it is Coprime no.

```
int a = 15;
int b = 8;
int gcd = 1;
for ( int i = 1 ; i <= Math.min (a,b) ; i++ ){
        if (a%i == 0 && b%i == 0){
            gcd = i;
        }
}
if (gcd == 1){
        Sop ("Coprime");
}else{
        Sop ("not coprime");
}
```

O/P = Coprime.

(n) check perfect no:

n = 6 divisors = 1, 2, 3 1+2+3 = 6.

if (sum of divisors == n) = perfect

n = 6 sum = 0.

i = 1 < 6

└ 6%1 == 0 sum = sum + i = 0 + 1 = 1

i = 2 < 6

└ 6%2 == 0 sum = 1 + 2 = 3

i = 3 < 6

└ 6%3 == 0 sum = 3 + 3 = 6

i = 4 < 6

└ 6%4 == 0 ✗

i = 5 < 6

└ 6%5 == 0 ✗

i = 6 < 6 — ✗.

Sum = 6 & num = 6.

So 6 is perfect no.

```
int n = 6;
int sum = 0;
for (int i = 1; i<n; i++) {
        if (n%i==0) {
                sum = sum+i;
        }
}
if (sum==n) {
        Sop ("perfect no");
} else {
        Sop ("not perfect");
}
```

O/P = perfect no.

TC = O(n)
SC = O(1)